



Measuring education: Do we need a plethora of assessment studies or just a single score?

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ARTICLE INFO

Keywords:

Large-scale assessment
Education
PISA
TIMSS
ICCS
ICILS

ABSTRACT

The major international large-scale assessments in education prioritize the three core domains such as mathematics, science, and reading, neglecting other important domains such as civic education or computer and information literacy. The present study aims at determining the added value of different international large-scale assessments to evaluating educational systems across the globe. Using country-level data from all recent international assessments by the IEA and the OECD, we conducted a series of correlational analyses and multidimensional scaling to determine the similarities and differences among educational assessments. The findings of this study show that the various measures of achievement correlate to different degrees at country level. While mathematics, science, and reading are the most closely correlated, the correlations with civic education and computer and information literacy are considerably lower. We conclude that subject domains as civic education and computer and information literacy add a significant value to the assessment of education.

Introduction

The most common approach to assess the quality of education is by measuring the knowledge and skills that students gained at school through standardized testing. In this regard, international large-scale assessments (ILSAs) provide indicators of educational quality and efficiently measure the educational systems in different countries. The first international assessments in education were administered in 1964 with then launch of the IEA's First International Mathematics Study (OECD, 2015), but the rise of ILSAs started since the mid-1990s (Addey et al., 2017), and since then, the number of studies, their frequency, and the number of participating countries have considerably increased.

Although the purpose of ILSAs which is to assess the scholastic performance of students is certainly worthy, there are particular concerns associated with the existing ILSAs. Thus, Emler et al. (2019) pointed out that the high cost is among the significant side effects of educational large-scale assessments since they require considerable finances to design, implement, and analyze the test studies. Countries not only have to cover the participation fees but also the costs to administer the surveys in the respective countries. Such high cost of ILSAs might be a burden for countries and their citizens as these expenses are covered by taxpayers.

Non-monetary costs for participation in ILSAs refer to the technical administration of testing and might present certain challenges for schools. Participation in large-scale assessments require selected schools to dedicate efforts for preparation and actual testing, including communication with ILSA staff, arrangement of pre-assessment visits to schools, and so on. On one hand, such involvement of schools can be characterized as capacity building process where school representatives acquire new competencies to administer testing. On the other hand, this experience can be time-consuming for teachers and school leaders as well as might divert their attention from other important duties to be performed at school. Furthermore, ILSAs are criticized for putting unnecessary pressure on students and teachers, distracting them from learning the curriculum and rather forcing them to prepare specifically for the international assessments.

Even though ILSAs provide valuable information, they are also associated with considerable costs. This cost-benefit tension related to the implementation and use of ILSAs leads to the questions about the necessity of administering a plethora of assessment studies in education. Do different ILSAs provide similar or disparate inferences about the educational systems across the globe? Should we focus on measuring a few subject domains such as mathematics, science, and reading or should we consider including other domains in ILSAs in order to obtain a

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<https://doi.org/10.1016/j.ijedro.2023.100281>

Received 11 February 2023; Received in revised form 19 July 2023; Accepted 31 August 2023

Available online 7 September 2023

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comprehensive picture of the education system? To answer these questions, this study aims at examining the correlations between various ILSAs and analysing their convergence and divergence.

Some may argue that it is sufficient to study only one subject domain (e.g., mathematics) to evaluate educational policy as several recent studies revealed that student performance tends to be similar across different domains (e.g., [Wendt & Kasper, 2016](#); [Bergold et al., 2017](#)). Other scholars also argue that the country scores of different ILSAs are typically quite similar ([Klieme, 2016](#); [Rindermann, 2007](#)) which challenges the idea that different ILSAs provide added value to inform policy and research. In the scientific literature, there is evidence that certain assessment studies, such as PISA in mathematics and TIMSS 8th Grade, show high similarity ([Klieme, 2016](#)); however, there is little evidence if the results of ILSAs in core domains (mathematics, science, and reading) and non-core domains (e.g., civic education, computer sciences, etc.) demonstrate substantial differences. If it turns out that there are considerable differences between test results in core and non-core domains, this may mean that some large-scale assessments are stronger correlated than others.

The overarching goal of assessment is to build a comprehensive picture of educational system which can be achieved by measuring multiple aspects of the system. Measuring the same domains or domains that are highly correlated is less likely to provide an added value. In this regard, a reasonable approach to address this issue could be to integrate the data from various assessments that measure different subjects, grades, and competences. This approach might provide valuable information about the educational progress and better guide researchers and policymakers in developing educational systems. The question remains about what aspects of the educational system can be tested in a way that it contributes to the further knowledge about education.

What and how do different ILSAs measure?

There are various international assessments in education that are conducted in specific regions and worldwide. Among the regional large-scale assessments, the most prominent ones are Latin American Regional Comparative and Explanatory Study (ERCE, for its Spanish name) and Southern African Consortium for Monitoring Educational Quality (SACMEQ) that focus on designing the achievement assessments in accordance with the regional educational context. As for the worldwide assessments, these are the international large-scale assessments (ILSAs) coordinated by the International Association for the Evaluation of Educational Achievement (IEA) and organization for Economic Cooperation and Development (OECD) that gained momentum in educational communities. Therefore, for the sake of simplicity, the focus of the present study is on the ILSAs which have been coordinated by IEA and OECD. These international assessments measure academic achievements of students and thus reveal cross-country variations as well as unique features of educational systems pertaining to a specific country ([Hanushek & Woessmann, 2011](#)). Moreover, the IEA- and OECD-led assessments provide representative samples of students and schools and in this way can capture different factors impacting the inputs, processes, and outputs of educational systems ([Gustafsson, 2008](#)).

Interestingly, the early IEA studies covered a wide range of domains to be tests in a single study which shows that education was viewed as an aggregation of multiple characteristics and measures which should be analysed in toto in order to evaluate the educational quality and progress. For example, the Six Subject Survey (1967–1971) comprised assessment in science, reading comprehension, literature, English as a foreign language, French as a foreign language, and civic education.

At present, the IEA ongoing studies, including the flagship TIMSS (Trends in International Mathematics and Science Study) and PIRLS (Progress in International Reading Literacy Study), are more focused on a narrow scope of thematic assessment. CIVED (Civic Education Study) represented a novel domain to test—civic education—and was conducted by IEA in 1999–2000 among students to assess their knowledge and

understanding of democracy, citizenship, diversity, and social cohesion. The recent innovations in ILSAs introduced by IEA are the International Civic and Citizenship Education Study (ICCS) and the International Computer and Information Literacy Study (ICILS) that were first conducted in 2009 and 2013 respectively. However, the number of countries participating in ICILS and ICCS is relatively small at least in comparison to the number of participating countries in TIMSS and PIRLS.

OECD has been spearheading the renowned programme for International Student Assessment (PISA) since 2000 to evaluate the scholastic performance of 15-year-old students worldwide. PISA primarily measures cognitive skills in three major subject domains: mathematics, science, and reading. Different cycles of PISA also included additional domains for assessment such as financial literacy, collaborative problem-solving, global competence. However, the additional domains in PISA are not obligatory for all participating countries, and typically the number of countries that have undertaken the assessment in additional subject realms is small compared to the full sample of countries in PISA.

[Table 1](#) presents an overview of the ILSAs conducted since 1995 till 2019, featuring the coordinating organization, target population, number of participating countries, and subject domains that were tested. In certain cases, the number of participating countries varies across tested domains within the same study and cycle. As can be seen from [Table 1](#), the majority of ILSAs have been conducted in the areas of mathematics, reading, and science, while other domains such as civic education or information literacy are tested less frequently and with fewer participating countries.

A possible explanation for the strong emphasis on dominant subjects among assessment studies can refer to the idea that the three domains (mathematics, science, and reading) are the strong predictors of labor force quality (e.g., [Chetty et al., 2011](#); [Dougherty, 2003](#); [Heckman et al., 2006](#)). This has prompted to use the academic achievement in these three domains as the primary measures in assessment studies (e.g., [Watts, 2020](#); [Chetty et al., 2011](#); [Curto & Fryer, 2014](#); [Deming, 2009](#); [Loeb et al., 2007](#)). However, [Chen and Luoh \(2010\)](#) challenged this idea based on their results of cross-country growth study which demonstrated that test scores in mathematics and science from PISA and TIMSS do not exhibit strong link to cross-country income differences.

Although the test items in PISA do not exclude curriculum-related aspects, they are rather based on the concept of competency and thus measure functional skills and the extent to which students are prepared to participate in the modern society ([OECD, 2003](#)). On the other hand, the distinct feature of many assessment studies administered by IEA is that they are designed in accordance with the official curriculum of participating countries ([Wagner & Hastedt, 2022](#)). The IEA have also conducted ILSAs that are not curriculum-based such as Pre-Primary Project, ICCS, and ICILS. Moreover, major ILSAs of both OECD and IEA encompass such domains as mathematics, science, and reading and largely focus on the population of approximately the same age group (for example, 15-year-olds in PISA and eighth-grade students in TIMSS, ICILS, and ICCS). Moreover, considering that OECD and IEA also intend to measure change in student achievement as well as country's educational environment in the long term, their ILSAs contain constructs and frameworks that tend to remain comparable across cycles within the programmes. Such a design of ILSAs allows for a longitudinal analysis to evaluate the changes in the context factors at both individual and school levels ([Jude & Kuger, 2018](#)).

The existing ILSAs test different subject areas, and the plurality of domains to be tested leads to the question of what learning outcomes are considered relevant and necessary for evaluating the effectiveness and efficiency of educational systems across countries over time. To understand this issue, we suggest examining various perspectives on education and the factors that shape student performance in ILSAs.

Table 1
International Large-Scale Assessments in Education 1995–2019.

Study	year	organization	Population	N ^a	Tested domains	Reading	Science	Problem-solving	Collaborative problem-solving	Financial literacy	Civic education	Computer and information literacy	Global Competence
					Mathematics								
TIMSS	1995	IEA	Grade 3, 4	26	x		x						
TIMSS	1995	IEA	Grade 7, 8	40	x		x						
TIMSS	1999	IEA	Grade 8	38	x		x						
CIVED	1999	IEA	Grade 8	27							x		
PISA	2000	OECD	15-year-old	43	x	x	x						
PIRLS	2001	IEA	Grade 4	36		x							
PISA	2003	OECD	15-year-old	41	x	x	x	x					
TIMSS	2003	IEA	Grade 4	29	x		x						
TIMSS	2003	IEA	Grade 8	51	x		x						
PISA	2006	OECD	15-year-old	57	x	x	x						
PIRLS	2006	IEA	Grade 4	47		x							
TIMSS	2007	IEA	Grade 4	44	x		x						
TIMSS	2007	IEA	Grade 8	57	x		x						
PISA	2009	OECD	15-year-old	73	x	x	x						
ICCS	2009	IEA	Grade 8	38							x		
PIRLS	2011	IEA	Grade 4	57		x							
TIMSS	2011	IEA	Grade 4	58	x		x						
TIMSS	2011	IEA	Grade 8	50	x		x						
PISA	2012	OECD	15-year-old	65	x		x	x ^b		x ^c			
ICILS	2013	IEA	Grade 8	21								x	
PISA	2015	OECD	15-year-old	73	x	x	x		x ^d	x ^e			
TIMSS	2015	IEA	Grade 4	56	x		x						
TIMSS	2015	IEA	Grade 8	46	x		x						
PIRLS	2016	IEA	Grade 4	57		x							
ICCS	2016	IEA	Grade 8	24							x		
PISA	2018	OECD	15-year-old	79	x	x	x						x ^f
ICILS	2018	IEA	Grade 8	14								x	
TIMSS	2019	IEA	Grade 4	64	x		x						
TIMSS	2019	IEA	Grade 8	46	x		x						

Note. (a) Number of participating countries and regions, including benchmarking entities; (b) $N = 44$; (c) $N = 18$; (d) $N = 17$; (e) $N = 54$; (f) $N = 27$.

Theoretical perspectives on education

The existence of a variety of student tests leads to the questions about the necessity of administering a plethora of assessment studies in education. Do different ISLAs provide similar or disparate inferences about the educational systems across the globe? One of the approaches to answer this question is to compare the country scores from different assessments. If countries perform similarly across the assessments, presumably there is little reason to participate in different assessments, and a single score in one domain could suffice to draw conclusions about the whole educational system of a country. Nevertheless, if countries perform differently on different assessments, then the more data we have from different assessments and subject domains, the more complete information about the educational systems we have. In that case, the question would be about which assessments and subject domains add more value to understanding the progress of educational systems.

There are several theoretical frameworks from economics, psychology, and sociology that could interpret the existing differences in test performance among students participating in ILSAs. This issue can be examined from the psychological perspective acknowledging the dominant role of the general intelligence in student achievement. As some proponents of this theory claim, different ILSAs mostly measure general intelligence, and this measuring approach largely contributes to similarity among different assessment studies. Given the effects of educational tracking on students' academic achievements, the social stratification viewpoint claims that the differences in test performance are due to the structure of school systems. Economic development argument sees the financial gap between countries as the major factor for unequal scores among participating countries.

Student general ability framing the scores in ILSAs

The general ability approach in ILSAs has traditionally been associated with cognitive psychology, specifically with the theory of general intelligence or ability (g-factor). It posits that students who perform well in one subject domain are more likely to achieve high scores in other subject domains. Rindermann (2007) claimed that ILSAs mostly measure the g-factor of students rather than their specific achievements in different domains. This explains the strong correlation between the performance of students in different subject domains in ILSAs. Based on these findings, Rindermann suggested that it is more appropriate to work with a sum values of different scales and studies in order to measure the quality and progress of education:

Research done at the national level about the effectiveness of education or characteristics of societies (economic growth, wealth, democracy, liberty, peace, health, etc.) should be done with sum values for a large sample of countries. The use of single scales or single studies is scientifically inappropriate. (p. 693)

However, it is crucial to acknowledge the need for a nuanced understanding of the associations between general ability and performance at both the individual and group levels (e.g., Härnqvist et al., 1994). At the country level, the focus is on the relationship between the overall performance of a country in ILSAs and various factors, such as socio-economic indicators, educational policies, or cultural influences. Such examination seeks to understand the broader educational landscape and identify factors that contribute to country-level performance differences.

On the other hand, studies focusing on the student level within a country explore the relationship between individual students' general ability and their performance in specific subject domains. These analyses aim to examine the extent to which general ability factors contribute to students' achievement within a specific educational context.

Therefore, it is imperative to differentiate between the associations observed at the macro level (country-level) and micro level (student-level) when discussing the influence of general ability on educational

outcomes in ILSAs. By recognizing this distinction, researchers can provide a more nuanced understanding of the complex interplay between general ability, subject-specific achievements, and broader educational factors at different levels of analysis.

Social stratification in education

Sociological theory proposes to consider differences in school structures when analysing differences in test results in international educational assessments (LeTendre et al., 2003; Kerckhoff, 2001). In countries with early tracking system, different curricula exist for different school tracks: Vocationally oriented students focus on learning the core domains such as reading, mathematics, and science, whereas academically oriented tracks cover a broader range of subjects. In contrast, countries with late tracking system offer the same curricula for all students despite their school track. These differences in schooling are expected to affect the test scores of students and eventually the country rankings in ILSAs.

Economic development and educational production function argument

Economic perspective regards student competencies or achievement as outputs of investments into school system (Hanushek, 1987). By increasing the resources invested in the school system, it is expected the student competencies can be enhanced which would lead to a better test performance of students in ILSAs. Therefore, a thorough analysis of test performance can reveal the preparedness and quality of the school systems worldwide (OECD, 2013). Countries with higher gross domestic product (GDP) per capita tend to perform higher in ILSAs (e.g., OECD, 2019) which supports the economic development argument claiming that the general level of resources allocated to support students have a direct impact on their scholar performance. Thus, countries with higher national incomes can allocate more funding to education sector, and this could explain their better performance in ILSAs in comparison with low-income countries.

Nevertheless, limited resources argument suggests that even in a group of similarly developed countries, there may be a variation in how each country spends the budget on education, either prioritizing a specific subject domain (e.g., mathematics) or a group of subjects (e.g., hard sciences). This may mean that students from developed countries might differ in their test performance depending on the investment priorities of their governments in the education sector.

Curricula around the world

The curriculum argument incorporates the above-mentioned theoretical perspectives: the curriculum is a reflection of the country's specific context, economic growth, and educational traditions which shape the student development and consequently their academic performance. National curricula may capture different things which would result in differences in the scores among countries in ILSAs.

The sine qua non for any ILSA is the comparability of the subjects to be measured across the participating countries. In this regard, the study by Benavot et al. (1991) provides a solid rationale for studying competencies internationally. Benavot et al. examined the educational curricula in mass school systems across countries and over an extensive period of time. The researchers analyzed the content of different national curricula and classified their subjects into general categories such as language, mathematics, natural sciences, social sciences, religious and moral education, esthetic education, physical education, health education, vocational education/practical subjects, and other. Benavot et al. (1991) asserted that, despite national variations in economic or political structures, the range of subjects that are taught in primary and secondary schools is virtually identical throughout the world. Based on the results of their study, Benavot et al. contented that the national school systems are shaped by the standardized models of mass

education. The main inference made is that the school curriculum is established at the global level which, in turn, provides ground for conducting ILSAs as they are based on the fundamental idea of comparability.

Reflecting on the balance between the subjects in the curricula, Meyer et al. (2017) stressed that the consistent trade-off can be seen not between language and mathematics but rather between language and science, arts, physical, and vocational education. This implies that certain countries give priority to language instruction, whereas other countries, on the contrary, put greater emphasis on a range of subjects other than language. Meyer et al. (2017) also reported that the modern European educational system has switched from the traditional language-laden curriculum to the curriculum oriented towards science, vocational training, and arts. However, Cha (2007) also made a comparative data analysis of the school content and suggested that foreign language instruction, namely English, has become an institutionalized subject at schools across the globe and an important locus in many national curricula.

Interconnectedness of different ILSAs

Much of the research in the realm of educational assessment is focused on the inter- and intra-dimensionality of test studies. With the rise of international educational assessment, there have been also a number of studies that compared different ILSAs.

Previous research has revealed high intercorrelations among different achievement tests. For instance, Bergold et al. (2017) noted that analyses of the PISA test scores in mathematics, reading, and science showed a strong intercorrelation at country level from $r = 0.75$ to $r = 0.81$ (Reilly, 2012) and even higher from $r = 0.95$ to $r = 0.99$ (Rindermann, 2007). TIMSS and PIRLS data from Grade 4 demonstrated lower but still significant correlations at student level in the range between $r = 0.54$ and $r = 0.74$ (Bos et al., 2012). It is important to note that correlations among various tests differ at student and country levels (Baumert et al., 2009). Some countries may perform high on all tests, while other countries may show low scores on all tests. Thus, if we look at the correlation at the country level, we might observe high correlation of all tests just because countries perform either high or low in all tests. However, at the student level within a country, tests may be not correlated or even negatively correlated. A student may be good at mathematics but poor at reading, and such individual scores would result in relative differences within countries. Rindermann (2007) compared the national test scores from PISA of 15-year-old student achievement and used adjustment formulas for students older or younger than the targeted age to compare TIMSS data of Grades 4 and 8, as well as PIRLS data from Grade 4 and concluded that the results are rather homogeneous. It was asserted that even though diverse scales of student assessment and diverse test approaches are used, different ILSAs measure the same construct. Rindermann (2007) suggested that the high correlations are due to the similarities of tests within studies, including cognitive demands for solving test items, as well as to the common developmental factors at the individual and national levels. Nevertheless, a caution must be applied when interpreting Rindermann's results as he used a set of highly heterogeneous countries for analysis. Even correlations of 0.8 and higher do not imply that all countries perform at the same level in various assessments. Anscombe's quartet illustrates that it is important to study data carefully because nearly identical data sets might have quite distinct distributions when graphed.

In contrast, Olsen and Grønmo (2006) compared TIMSS and PISA and noted that some Nordic and English-speaking countries performed relatively better in PISA than in TIMSS, while Eastern European countries performed relatively better in TIMSS. They explained these findings by the differences in methodologies that the two studies apply. Thus, the researchers pointed out that the assessment frameworks in these two ILSAs are constructed following different perspectives on mathematics. Moreover, the items in PISA test the competence of using the numerical

information taken from real world context, while TIMSS focuses on pure mathematics and comprises items which require knowledge from the field of algebra and geometry (Olsen & Grønmo, 2006).

Wu (2010) compared the similarities and differences between PISA 2003 and TIMSS 2003. She utilized several parameters for an in-depth comparison, scrutinizing survey methodologies, mathematics frameworks, and content of the tests. Echoing the findings by Olsen and Grønmo (2006), Wu claimed that Western countries performed relatively better in PISA than in TIMSS; however, Asian and Eastern European countries demonstrated better scores in TIMSS as compared to their results in PISA. Wu linked these differences between countries' performances in two ILSAs to the content balance of the tests and the sampling approaches, namely pointing out that TIMSS adopts grade-based criterion for sampling, whereas PISA is guided by the age-based selection of students.

The present study

The existing assessment studies in education address different learning outcomes in education as well as follow specific assessment frameworks and methodological procedures. However, researchers pointed out that the results of large-scale educational assessments are rather homogenous and thus provide similar inferences about the academic performance of test takers. Consequently, having such a variety of assessment studies raises the concern whether they duplicate each other in terms of their input, or they bring different insights into the quality of educational systems worldwide.

Previous research on comparability of ILSAs primarily examined core subject-domains such as mathematics, science, and reading, leaving out other important domains such as civic knowledge and computer literacy. Moreover, most studies were limited to a subset of countries or regions to be included in the correlational analysis of ILSAs. These limitations of previous research make their findings incomplete, and it still appears to be problematic to answer the question whether different ILSAs provide added value to assess the quality of education.

The present study investigates the necessity of having such a variety of assessment studies as well as educational indicators and aims at shedding light on the interconnectedness between them. In particular, the study focuses on various ILSAs, which have been administered by OECD and IEA worldwide, and examines the correlations between them from the perspectives of the content and sampling approach.

The hypotheses that underpin this study are formulated below.

- H₁: The correlation is higher for different studies measuring the same subject domains than for different studies measuring different subject domains.
- H₂: The correlation is higher for studies measuring the same age cohort in comparison with studies of different age cohorts.
- H₃: The correlation is higher for studies closer in time than for studies that are more distant in time.
- H₄: The correlation is higher for studies administered by the same organization than for studies administered by different organizations (IEA vs. OECD).

Methods

Sample

The study used the data from different cycles of PISA, TIMSS, ICILS, and ICCS. The ILSAs selected for the analysis covered such domains as mathematics, reading, science, problem-solving, collaborative problem-solving, financial literacy, civic education, computer and information literacy. The number of participating countries/regions might vary depending on the cycle of the study as well as on the tested domain. The analyses we conducted involved all countries that participated in the selected ILSAs. However, such analysis of the data of all participating

countries combines diverse countries ranging from developing to highly developed. To address this issue, we replicated the analysis with a subset of OECD countries as a robustness check. While these analyses reduce the number of countries, the results remain qualitatively the same (see Appendices A.1, A.2, A.3, A.4, and A.5). The criteria for selecting the subgroup of OECD countries are the common characteristics of these member states, such as democratic governance and high-income economy. Although each of the selected country has its own distinct education system, the above-mentioned common characteristics of the countries are expected to contribute to the more or less homogeneous educational outcomes.

Measures

Achievement scores

The achievement variables are all the domains that were included in the different cycles of the ILSAs. All cycles of PISA tested mathematics, science, and reading, while additional domains were included in certain cycles, for example, global competence in PISA 2018. TIMSS consistently evaluated the student performance in two domains: mathematics and science. The relatively new studies ICILS and ICCS focus specifically on computer and information literacy and civic knowledge respectively.

The test scores of each International Large-Scale Assessment (ILSA) are comparable within the different study cycles but not across different studies. Within each ILSA, the achievement scales were designed to have an average score of 500 and a standard deviation of 100 among the participating countries in the first study cycle. In subsequent study cycles, the same scales are used, but the means and standard deviations may differ due to changes in the sample of countries and progress in national performance distribution over time. In PISA, the scale for achievement scales were established in the year when a particular domain became the primary focus for the first time (e.g., reading in 2000, math in 2003). It is important to note that the metric of the PISA scale is based only on OECD countries. Moreover, it should be noted that the achievement scores TIMSS have different scales for Grade 4 and Grade 8 assessments." **Analyses**

Correlational analysis

The first methodological step was to conduct the correlational analysis of the country mean scores based on the data of the selected ILSAs. The analysis intended to compare the mean scores of different countries in two subject domains which were tested in ILSAs comparably at the same time. For this purpose, TIMSS 2019 Mathematics was compared with TIMSS 2019 Science for both Grades 4 and 8, demonstrating the comparison between two different subject domains tested within the same assessment study, as well as TIMSS 2019 Mathematics was correlated with ICCS 2016 for Grade 8, showing the comparison between two different subject domains tested within different assessment studies. These studies were selected for the correlational analysis because we aimed at demonstrating whether country scores in different subject domains equate and provide the same information about the educational achievement of a country.

Furthermore, we compared the correlations between studies testing the performance of different age cohorts. We used the data from the six cycles (1995, 2003, 2007, 2011, 2015, 2019) of TIMSS Grade 4 and Grade 8 and compared how the performance in mathematics and science of fourth-grade students are correlated with the performance of eighth-grade students.

To illustrate country test performance over time, we computed correlations for countries participated in PISA reading literacy assessment among 15-year-olds starting from 2000 till 2018, using the latest PISA 2018 as a reference point. This correlation was done to test our hypothesis on whether studies that are closer in time show higher correlation in comparison to studies that are more distant in time.

Multidimensional scaling

In the next step, the multidimensional scaling was applied. Multidimensional scaling (MDS) represents a non-linear dimensionality reduction strategy to visualize the level of similarity between data by displaying the distance matrix (Cox & Cox, 2008). We conducted MDS by computing cross-country correlations in order to reduce the complexity of the data analysis. The multidimensional scaling was applied to analyze the latest cycles of PISA, TIMSS, ICCS, and ICILS and visualize how these studies are correlated to each other.

Results and discussion

Correlational comparison of different subject domains

The correlational analysis of the country mean scores from ILSAs only partially support our assumption that assessment studies produce different indication of how well students perform on tests from a particular country. Fig. 1 demonstrates a high correlation of 0.94 between the performance of participating countries in TIMSS 2019 Science and Math for Grades 4 and 8. We can see here that countries perform relatively similarly in these two domains within the same assessment. Hence, based on this graph, we can assert that high performance in mathematics is a good predictor of high performance in science. In other words, there is little added value of a science score if we already have a math score – and vice versa. Fig. 1 shows that the performance of countries in two subject domains in TIMSS gives the same information about the progress of educational system as the correlation is rather high between two domains.

The correlational comparison of different subject domains from ILSAs was conducted also for 34 OECD countries as illustrated in (Fig. 2). The correlational comparison of different subject domains from ILSAs was conducted for OECD countries as illustrated in Fig. 2. Consistent with the pattern observed for all participating countries, the findings for OECD countries reveal the following:

1. TIMSS 2019 mathematics and science scores exhibit a high correlation, indicating a strong positive relationship between these test domains.

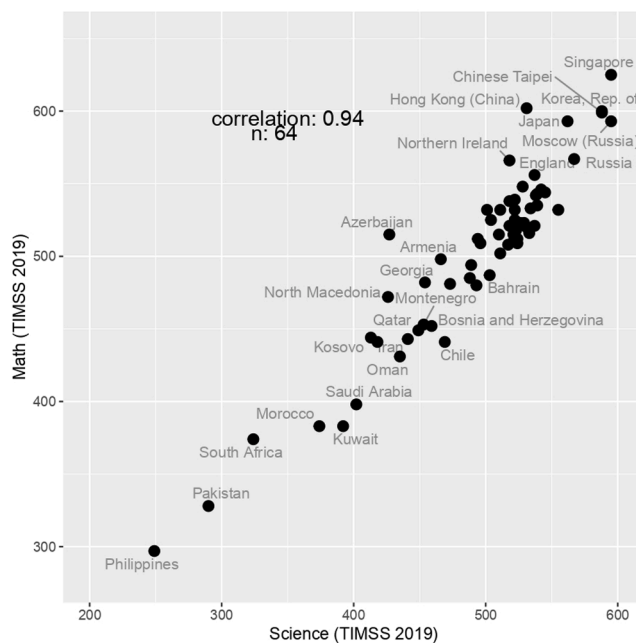


Fig. 1. Country mean performance in TIMSS 2019 Mathematics and Science for Grades 4 and 8.

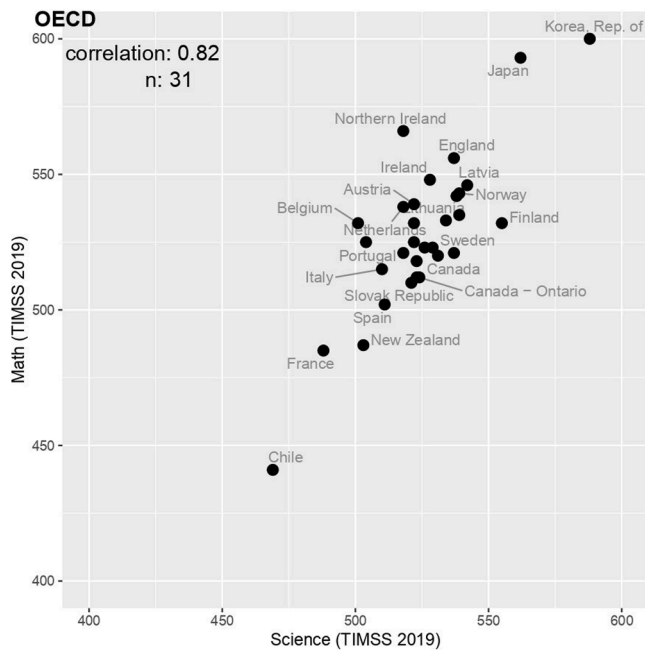


Fig. 2. Mean performance of OECD countries in TIMSS 2019 Mathematics and Science for Grades 4 and 8.

2. This strong correlation suggests a parallel performance in mathematics and science among OECD countries, emphasizing the interconnectedness of these academic domains.

These findings confirm the coherence and consistency in educational outcomes related to mathematics and science across the OECD countries.

In Fig. 3, the country mean performance in TIMSS 2019 Mathematics is compared with the results in ICCS 2016. It should be mentioned that the correlation in Fig. 3 is driven by several outliers which are Hong Kong, Republic of Korea, and Chinese Taipei at the top and Chile at the bottom. Here, conspicuous contrast in the test scores between these two

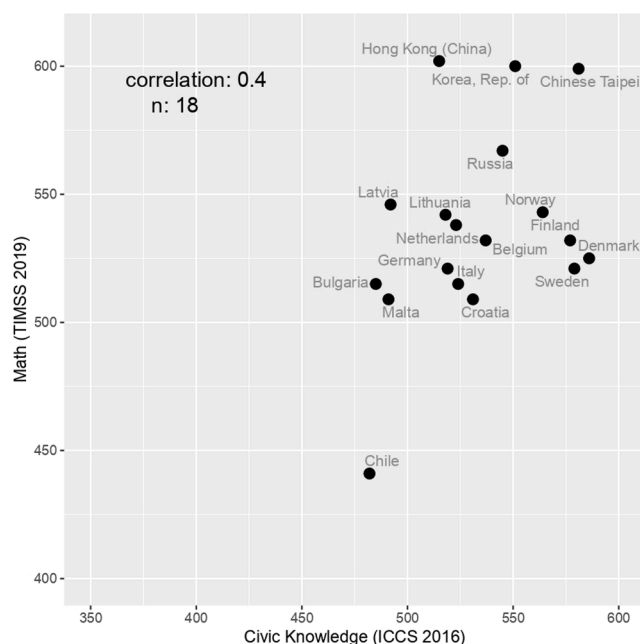


Fig. 3. Country mean performance in TIMSS 2019 Mathematics and ICCS 2016.

assessments can be found in the performance of certain countries. For instance, Latvia, Lithuania, and Norway have approximately the same score in TIMSS math: Latvia scored 546, Norway – 543, and Lithuania – 542. However, their national mean performance is utterly disparate in ICCS 2016, placing some countries far away from the countries with which they shared similar test scores in TIMSS, and thus giving a completely different picture of educational achievement. In ICCS 2016, Latvia, Lithuania, and Norway received 492, 518, and 564 scores, respectively. Thus, the gap between Latvia and Norway is 72 points in ICCS which corresponds to more than two schooling year (Willms, 2004a, 2004b). In this case, there is a true added value of having additional information about the educational system of a country.

The correlational analysis was conducted also for OECD countries that took part in the ILSAs in question (see Fig. 4). It was revealed that the correlation pattern was similar in the analyses involving OECD and all participating countries. This finding indicates that the pattern and relationship among ILSAs remain stable regardless of the selected sample.

Comparing different age cohorts

The comparison between different age cohorts gives additional findings about the added value that ILSAs bring to the educational community. To analyze how ILSAs measuring different age groups are related to each other, we took data from TIMSS Mathematics and Science Grade 4 and Grade 8 in the timeframe between 1995 and 2019 (see Fig. 5). Interestingly, two cohorts perform relatively similarly in math, but their performance in science fluctuates over the time. A high correlation in mathematics suggests that countries that do well in Grade 8 were already good in Grade 4. By implication, the added value is low for TIMSS Math, and the inferences about the performance of one of the grades can be predicted based on the performance of another grade. On the other hand, the mediocre and low correlation in science implies that there is an added value to conduct assessment in this subject domain in two grades. Moreover, further analysis could demonstrate whether a country does well in science in primary school and poor in secondary school or vice versa. Such an analysis might help a country to focus on investing more in science field at the relevant educational level.

To check the robustness of the correlations of the sample of all

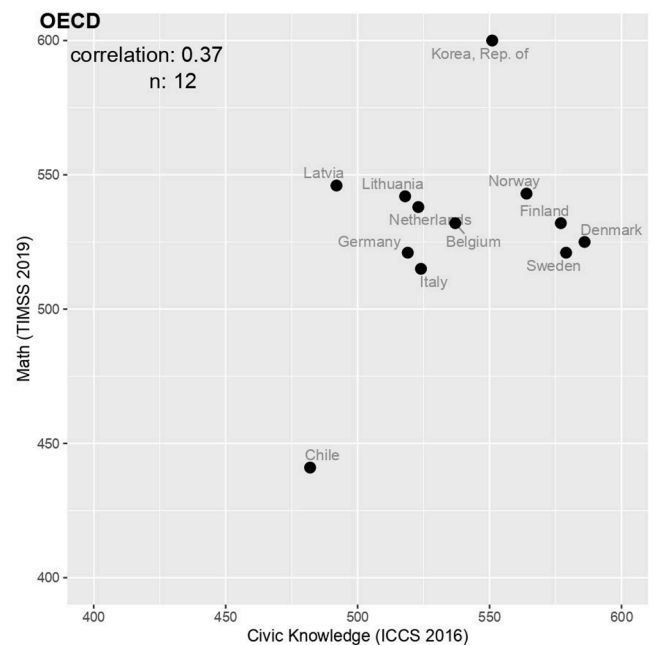


Fig. 4. Mean performance of OECD countries in TIMSS 2019 Mathematics and ICCS 2016.

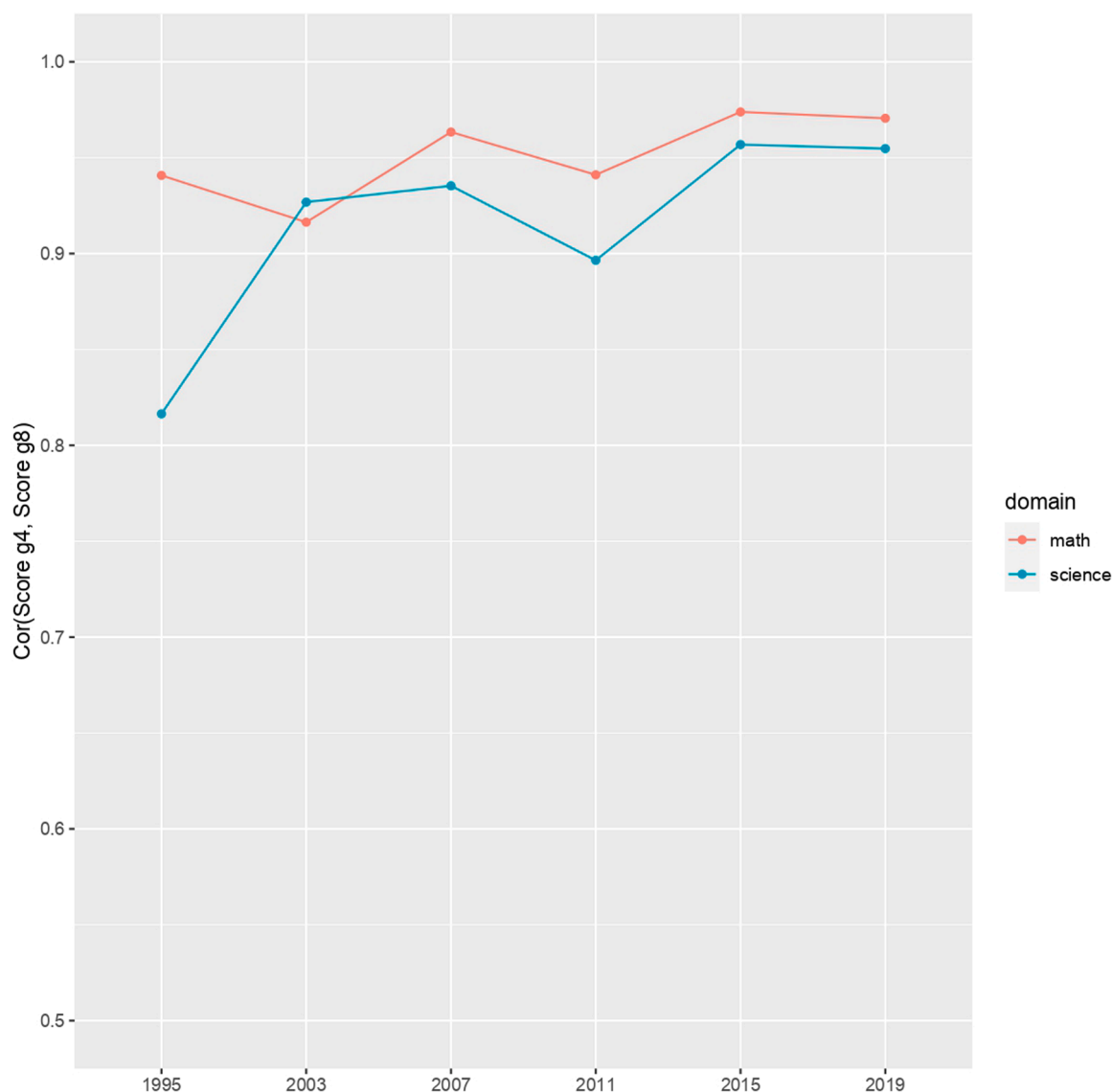


Fig. 5. Correlation between TIMSS Science and Mathematics Grade 4 and Grade 8, cycles 1995–2019.

countries, we conducted the same correlational analysis for OECD countries (see Fig. 6). The results of the analysis demonstrated that the dynamics of correlations between TIMSS Science and Mathematics of fourth- and eighth-graders are similar for the OECD and all countries: math performance is highly correlated; however, the performance in science varies over the years.

Changes over time in country performance

Fig. 7 shows the country mean scores in PISA reading literacy over time, starting from 2000 and till the latest cycle in 2018. Initially, we assumed that the correlation would be higher for studies closer in time than for studies that are more distance from each other in time. However, we do not observe this pattern in our findings. Although the correlation appears to be higher for studies closer in time for certain countries such as Chile, Estonia, the Netherlands, Poland; this is not a general pattern that can be identified here. Hence, our hypothesis concerning the correlation among ILSAs over time is not supported.

It is also important to acknowledge the impact of the weak test design in PISA until 2012, which introduced relatively large linking errors that affected the comparability of results across different assessments (Robitzsch & Lüdtke, 2019). The presence of these linking errors

can lead to substantial variability in country reading mean scores and may have influenced our findings. Thus, considering the limitations imposed by the test design during that period is necessary when interpreting our study's results and understanding the potential sources of variability in country-level performance.

Multidimensional scaling of the ILSAs

We applied the multidimensional scaling to compare systematically all the available data from the recent cycles of different ILSAs: PISA 2018, TIMSS 2019, ICCS 2016, and ICILS 2018. First, we present the heatmap in Fig. 8 which demonstrates the level of correlations among the selected ILSAs along with the number of countries that took part in both assessments.

In Fig. 8, it is evident that some assessments are more correlated than others. We can see here that test domains in PISA are highly correlated as well as the two domains in TIMSS which are also correlated to a greater extent. Moreover, Fig. 8 reveals that the number of countries that participated in tests on core domains such as math, science, and reading is much higher in comparison to the number of countries participating in tests on non-core domains, for example, global competence, computer and information literacy, civic knowledge. It can be inferred that all

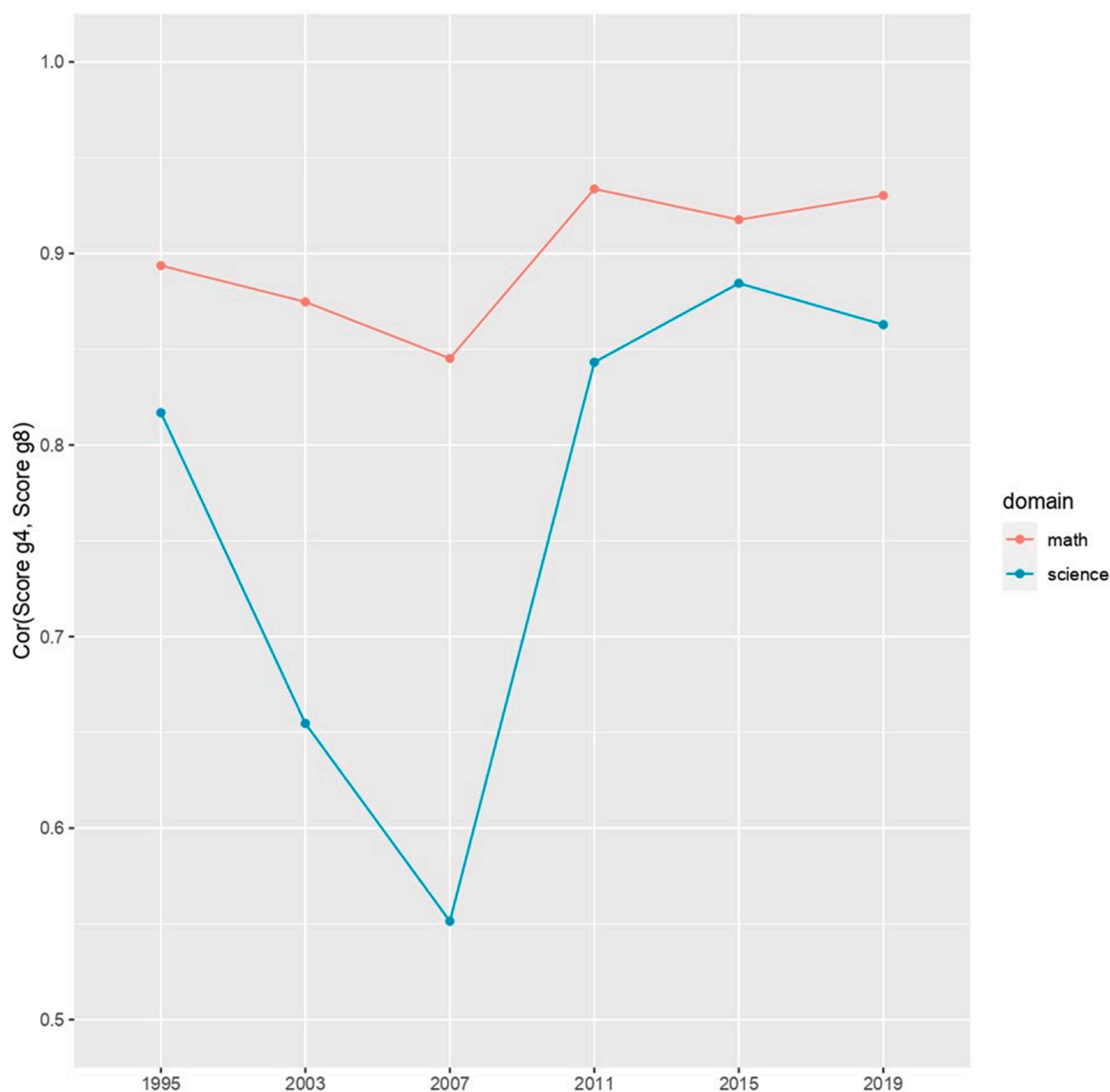


Fig. 6. Correlation between TIMSS Science and Mathematics Grade 4 and Grade 8 in OECD countries, cycles 1995–2019.

participating countries have information on how their students perform in reading, science, and mathematics in PISA, but only one out of three has information on how well students perform in civic education. In the most extreme case, only two countries have information on global competence in PISA and computer literacy in ICILS. Interestingly, PISA reading competence is more highly correlated with ICCS and ICILS than with PISA mathematics or science.

In Fig. 9, we present a heatmap specifically focused on OECD countries, further highlighting the interrelationships among the selected ILSAs: PISA 2018, TIMSS 2019, ICCS 2016, and ICILS 2018. Similar to the overall heatmap in Figs. 8, 9 reveals varying levels of correlations among the assessments for OECD countries. The domains within PISA exhibit stronger correlations with each other and with ICCS and ICILS compared to TIMSS mathematics or science. Additionally, we observe a similar pattern to the overall heatmap, wherein the number of participating countries varies across the core and non-core domains. Specifically, more OECD countries have data available on reading, science, and mathematics performance, whereas fewer countries have information on non-core domains such as global competence and computer literacy.

To ease the interpretation of the heatmap, we used multidimensional scaling to reduce the complexity of the data. This is also one approach to

address the issue that bivariate correlations are often based on very few countries only. Fig. 10 shows how different ILSAs are correlated and can be clustered together. Thus, we can clearly identify three clusters which are characterized by the thematic domain or by the coordinating institution. For example, PISA test domains coordinated by OECD can be clustered in one group as they demonstrate high correlations among themselves. Similarly, TIMSS mathematics and science studies administered by IEA are clustered in another group. Assessments on minor domains such as global competence, civic knowledge and computer and information literacy can be clustered in the third group and can be referred to the thematic grouping as these studies belong to the non-core domains.

Hence, Fig. 10 supports two hypotheses stated in the present study. It shows evidence that those studies which are conducted by one organization are higher correlated: PISA studies by OECD show higher correlations among each other, and IEA's studies as well. We had also another hypothesis that correlation is higher for studies measuring the same subject domain than for studies measuring different subject domains. In Fig. 10, we can see that TIMSS and PISA mathematics and science are placed in one cluster, and we can see another cluster of studies measuring non-core domains. Despite the issues related to the sometimes small samples, the multidimensional scaling reveals a pattern

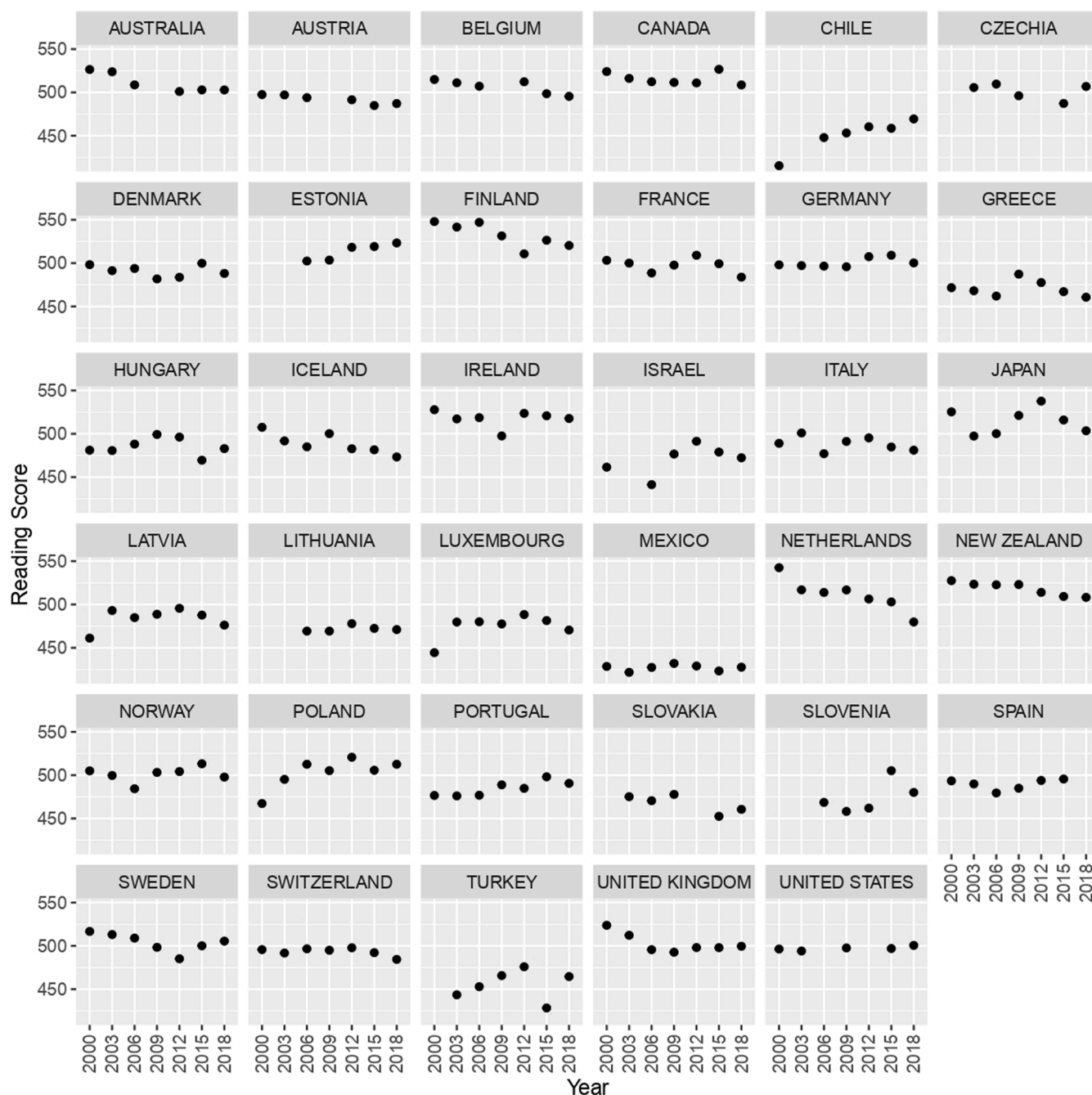


Fig. 7. Correlation between PISA country mean scores in reading performance and assessment years (2000–2018).

which we can interpret surprisingly well in substantive terms.

To check the results of this analysis based on sample of all participating countries, we also replicated the correlational matrix and multidimensional scaling for OECD countries participated in the selected ILSAs (see Fig. 11). The analysis for OECD countries shows the same pattern of correlations of ILSAs.

Conclusions

The education as well as school curricula around the globe encompasses a wide range of subjects, yet the ILSAs mostly focus on measuring the scholastic performance in mathematics, science, and reading. The present study set out with the aim to analyze the interrelation of ILSAs and identify the value they add to the comprehensive assessment of educational system based on the similarity or divergence of conclusions

one can draw relying on the test scores from different ILSAs. Our results showed that there are conspicuous differences between country mean performance in various ILSAs which leads to contrasting interpretations about the national educational achievement. Such disparity was evident through the analysis of ILSAs which included different subject domains and coordinating institutions. The gap between the country mean performance in different ILSAs was particularly prominent when the analysis involved ICCS and ICILS. This finding indicates that the assessments on civic knowledge and computer and information literacy are more likely to alter the traditional perspective on the national achievement which is formulated based on the country performance in the core domains such as mathematics, science, and reading.

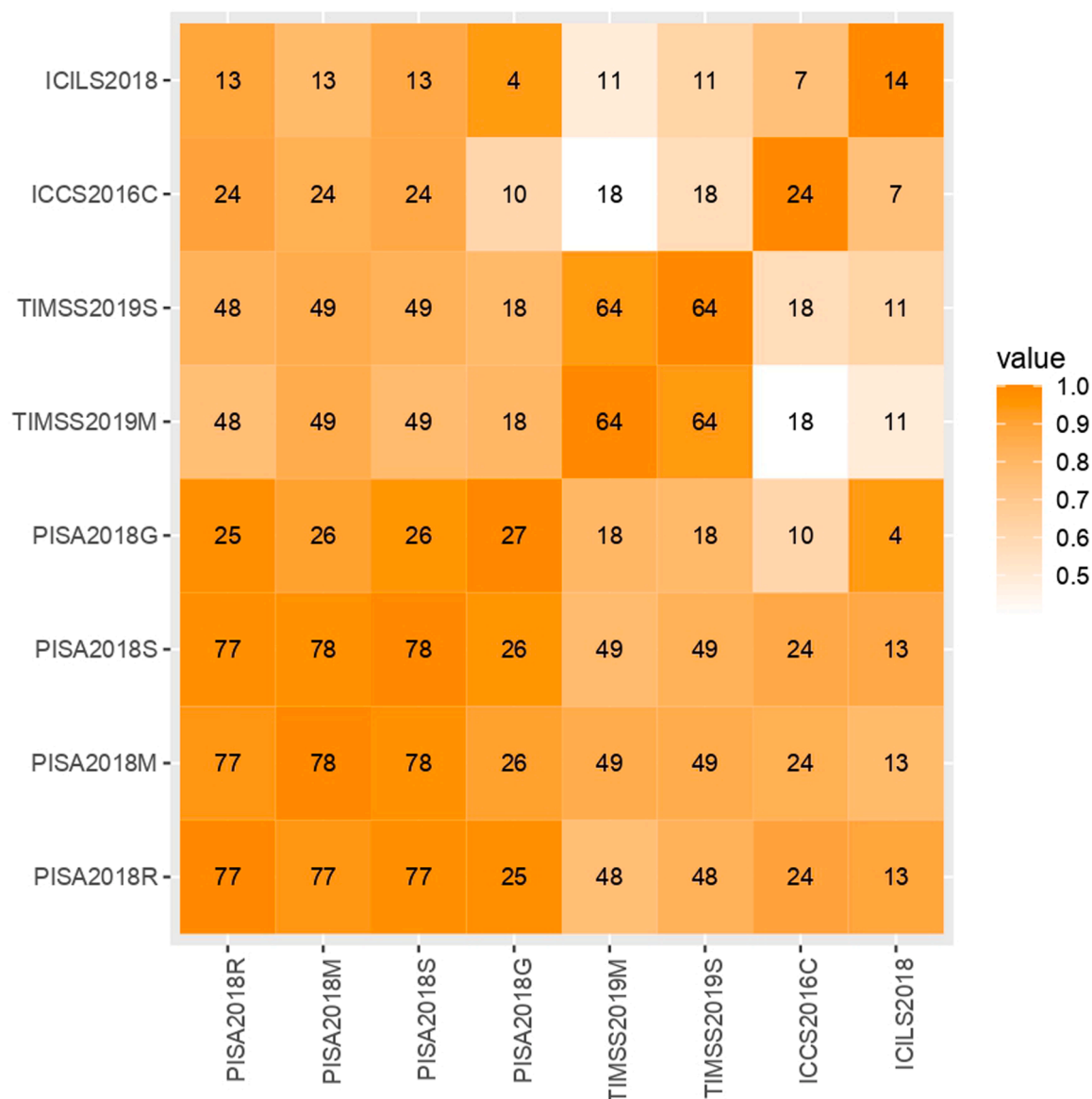


Fig. 8. Heatmap of correlations among PISA 2018, TIMSS 2019, ICCS 2016, and ICILS 2018.

Importance of participation in non-dominant studies

Unfortunately, there is a small number of countries that participate in the tests in those subject domains that bring additional benefits to the educational assessment. It can be assumed that national authorities might have the preconceived idea that tests in mathematics, science, and reading can fully capture the progress and quality of their educational systems. The results of our study challenge this view on educational assessment and demonstrate the importance of participating in less popular but still highly relevant test domains. Thus, it can be advised to countries and educational policymakers to consider participating not only in the largest assessment studies such as PISA and TIMSS, but also in smaller-scale studies such as ICCS and ICILS that have the largest added value.

Emphasis on a limited set of subjects

Among existing ILSAs, there is a strong emphasis on mathematics, science, and reading in terms of the number of assessment studies as well as the number of participating countries. More countries tend to actively participate in studies that measure the dominant subjects such as

mathematics, science, and reading, while other studies testing non-dominant subjects involve much fewer countries.

Taking a more holistic perspective in measuring learning outcomes is possible. Benavot et al. (1991) has shown that countries not only have in common subjects such as mathematics, science, and reading, but subjects such as physical education, music, fine arts, history and others are also well represented in the curricula around the world. Importantly, it is possible to test a wider range of subjects through assessment studies. For example, the Six Subject Survey has tested six subject domains in the 1970s.

The narrow focus on mathematics, science, and reading has implications not only for policy makers but also for researchers. ILSAs are increasingly used for secondary analyses (Hopfenbeck et al., 2018; Lenkeit et al., 2015), and for this reason ILSAs might set a misleading research agenda if they fail to consider other educational outcomes.

Limitations of the study

The present study examines the academic performance in ILSAs at the country level, which goes beyond the individual test performance and consequently shows the average national score in the specific

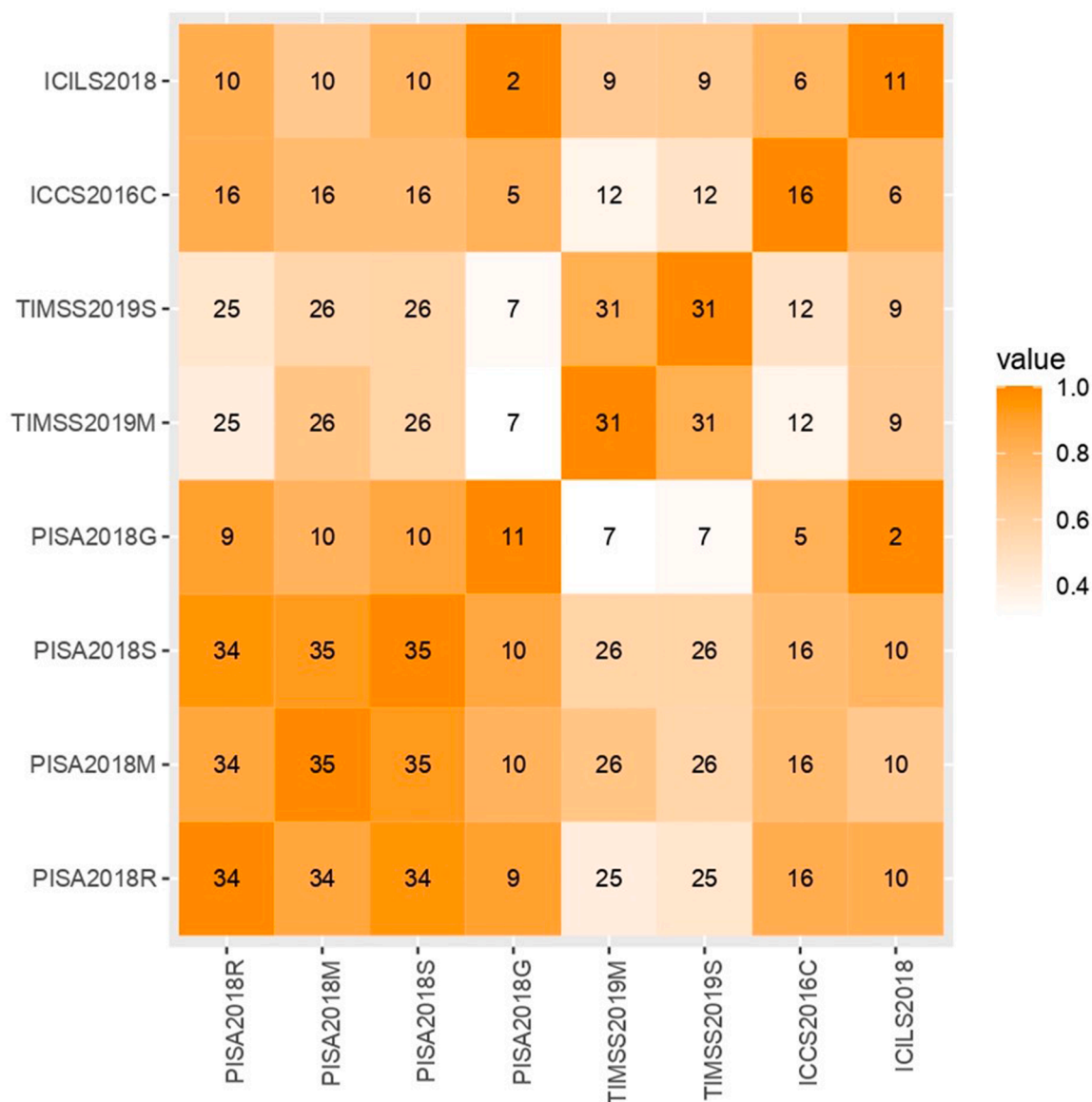


Fig. 9. Heatmap of correlations among PISA 2018, TIMSS 2019, ICCS 2016, and ICILS 2018.

assessment. The data sets from ILSAs contain countries that did not meet the sampling criteria or that nearly met these criteria after replacement of schools. We included such countries/regions in the analysis in order to build a more comprehensive picture of how different assessments might demonstrate dissimilar conclusions about educational systems. However, we admit that such analysis has limitations in its validity and reliability due to practical constraints of the available data. On the other hand, if the national data that do not correspond to the sampling criteria were excluded from the analysis, then the number of countries would be insufficient to draw comparisons and conclusions. The small sample is a natural limitation in international comparative research, and hence, further ILSA data collection with sufficient number of participating countries is required to determine reliable comparison between educational assessments.

The focus of the present study was on academic achievement of students, although there are other important educational outcomes that can be measured and analysed, for instance, attitudes, motivation, and values of students.

In this paper, we examined the comparisons of student scores in ISLAS at the country level. We neither argue nor think that our findings can be generalized on the student-level analysis. Both perspectives on

academic achievement are relevant for analysis; however, we decided to focus on country-level analysis because multiple measures are available on country but not student level: We included in the analyses those countries that took part in various ILSAs, but there are different independent samples of students that participated in different ILSAs. This makes it more challenging to draw comparisons of different ILSAs achievement at the student level.

Funding

This project has received funding from the European Union's Framework Programme for Research and Innovation Horizon 2020 (2014–2020) under the Marie Skłodowska-Curie Grant Agreement No. 765400.

Availability of data and material

The data that were used in this study are available at <https://timss.bc.edu/timss1995i/Database.html> (TIMSS 1995 International Database), <https://timssandpirls.bc.edu/timss2003i/userguide.html> (TIMSS 2003 International Database), <https://timssandpirls.bc.edu/timss2007/id>

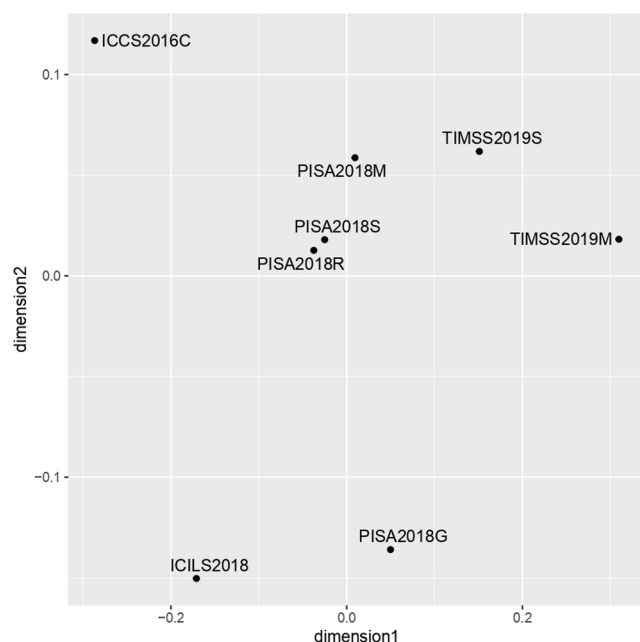


Fig. 10. Multidimensional scaling of data from PISA 2018, TIMSS 2019, ICCS 2016, and ICILS 2018.

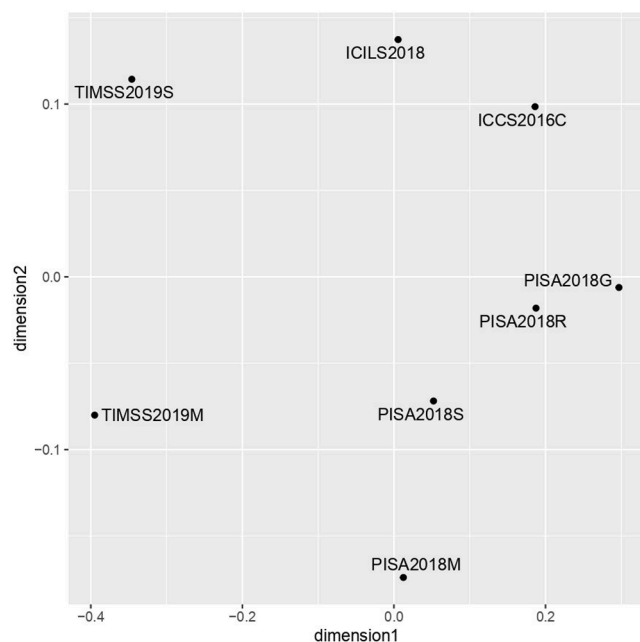


Fig. 11. Multidimensional scaling of data from PISA 2018, TIMSS 2019, ICCS 2016, and ICILS 2018 for OECD countries.

b_ug.html (TIMSS 2007 International Database), <https://timssandpirls.bc.edu/timss2011/international-database.html> (TIMSS 2011 International Database), <https://timssandpirls.bc.edu/timss2015/international-database/> (TIMSS 2015 International Database), <https://timss2019.org/international-database> (TIMSS 2019 International Database), <https://www.oecd.org/pisa/data/> (PISA 2000, 2003, 2006, 2009, 2012, 2015, 2018 International Database), <https://www.iea.nl/data-tools/repository/icils> (ICILS 2018 International Database), <https://www.iea.nl/data-tools/repository/iccs> (ICCS 2016 International Database).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors thank Robin Grugel for his continuous support in data analysis for this study.

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